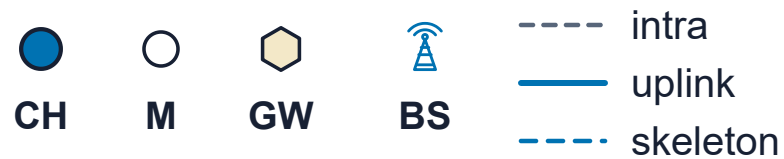
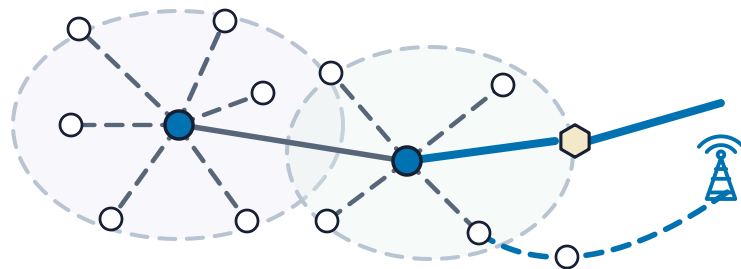


AERIS round

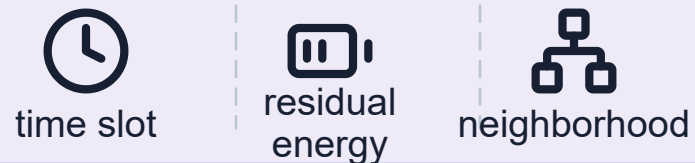


Phase 1 Cluster Formation



1

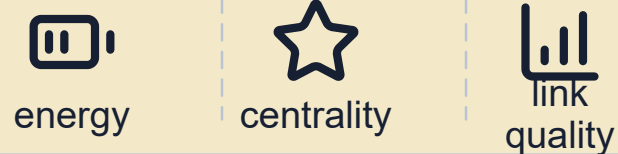
Round state



2

CH scoring

Elect CH



3

Cluster association



Phase 2 Intra-Cluster Communication

Inputs: energy · link quality · density · buffer

1

CAS mode selector

select one mode

Direct

M → CH

○ → ●

Chain

M → M → CH

○ → ○ → ●

Two-hop

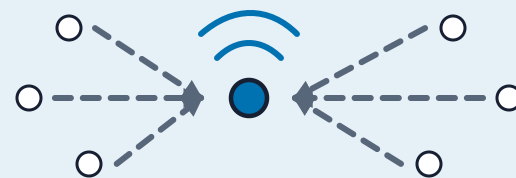
M → R → CH

○ → ○ → ●

○ M ⊕ R ● CH

2

Intra-cluster forwarding



3

CH aggregation



deduplicate



fuse



buffer



Phase 3 Uplink to BS

pre-transmission route selection

Q1

Direct acceptable? Yes

Direct uplink



CH → BS

No

Q2

Gateway valid? Yes

Gateway-assisted uplink



CH → GW → BS

No

Q3

Skeleton path valid? Yes

Skeleton reserve



CH → CH → BS

No

One-shot fallback



CH → BS

single attempt